POS Database Schema

Tables, columns, relationships and ERD diagrams of the pos-backend PostgreSQL database

18	31	12	202
tables	foreign keys	enums	columns

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Generated 2026-10-30 from the live development database (PostgreSQL, TypeORM migrations up to DeviceMomoCode1793220000000).

Overview

Areas

- **Company**: companies
- **Users and access**: users, employees, employee_roles
- **Catalog**: products, product_categories, files
- **Sales**: carts, items, orders, payments
- **Floor plan**: floor_maps, rooms, room_types, seats
- **Devices**: devices, device_activities
- **History**: audit_logs

How a sale moves through the tables

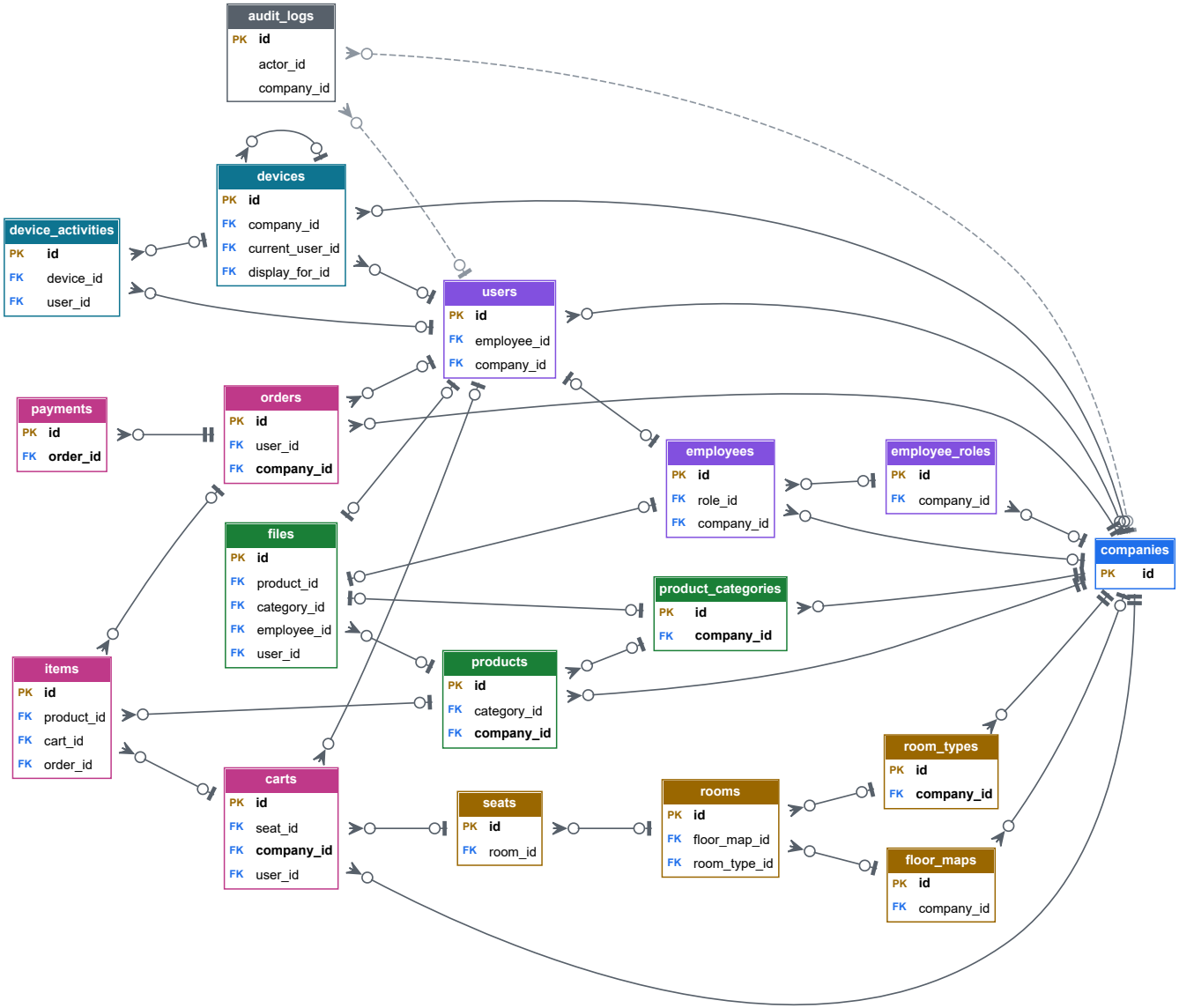
1. A worker signs in on a `devices` row (POS terminal) with their `users` PIN.
2. A `carts` row is opened (optionally for a `seats` row).
3. Each tap or scan adds an `items` row pointing to a `products` row.
4. Checkout creates an `orders` row, moves the items to it and deletes the cart.
5. The kitchen moves the order to PREPARING / READY; the till adds `payments` and completes it.
6. Creates, changes and deletes along the way are recorded in `audit_logs`.

Conventions

- Every table has a `uuid` primary key `id` (generated).
- Most tables have `createdAt` / `updatedAt` and `createdBy` / `updatedBy` (JSON: id, name, role of who did it). They are left out of the diagrams to keep them readable, but listed in the table reference.
- Data is separated per company: business tables carry `company_id` directly or through their parent (items → cart/order, payments → order, rooms → floor map, seats → room).
- Money is `numeric` (exact), never floating point.
- Names, emails, phones and barcodes are unique per company, not globally (except `users.email` and `companies.code`).
- Deleting a company cascades to its carts, orders, products, categories and room types; other links are restricted or set to NULL as listed under Relationships.
- The `migrations` table (TypeORM bookkeeping) is not shown.

Full ERD

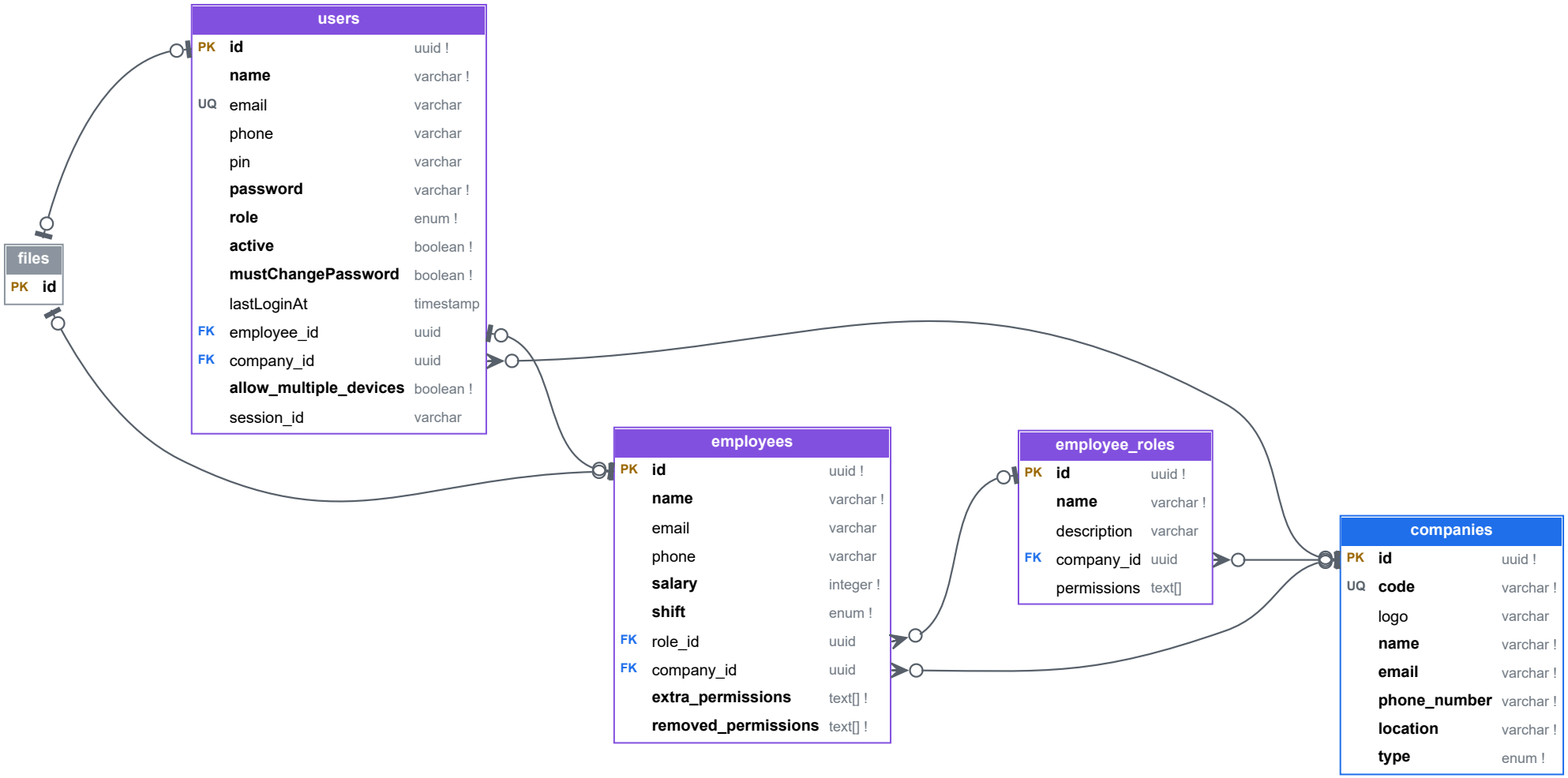
All tables with their keys (PK / FK columns only). Each line goes from the foreign key to the row it points to.



PK primary key · FK foreign key · UQ unique · **bold** / ! = required (NOT NULL) · exactly one · zero or one · zero or many · by id only (no FK)

ERD: Company, users and access

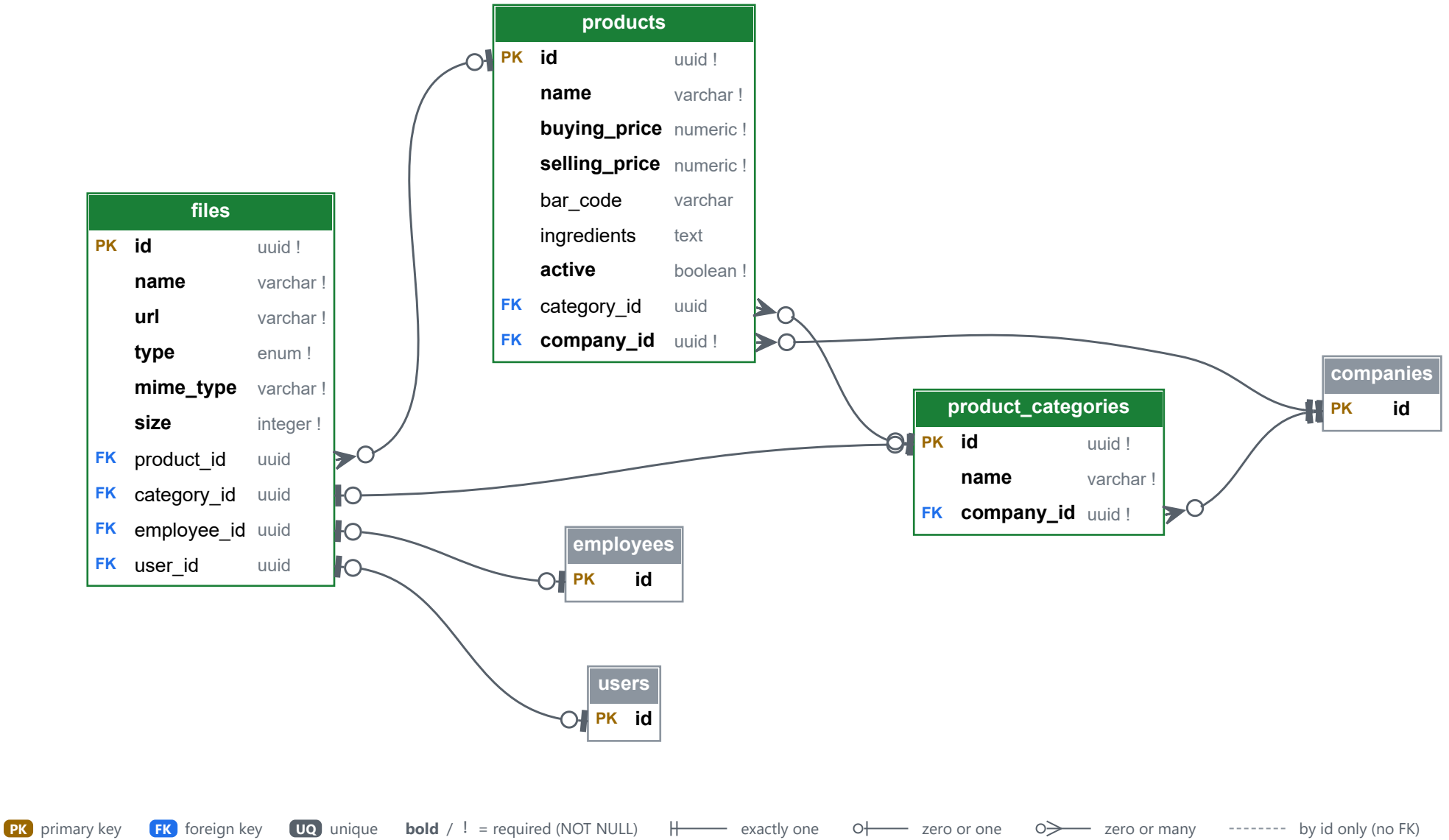
Who works for a company and what they are allowed to do. Grey tables belong to another area (only their id is shown).



PK primary key · FK foreign key · UQ unique · **bold** / ! = required (NOT NULL) · exactly one · zero or one · zero or many · by id only (no FK)

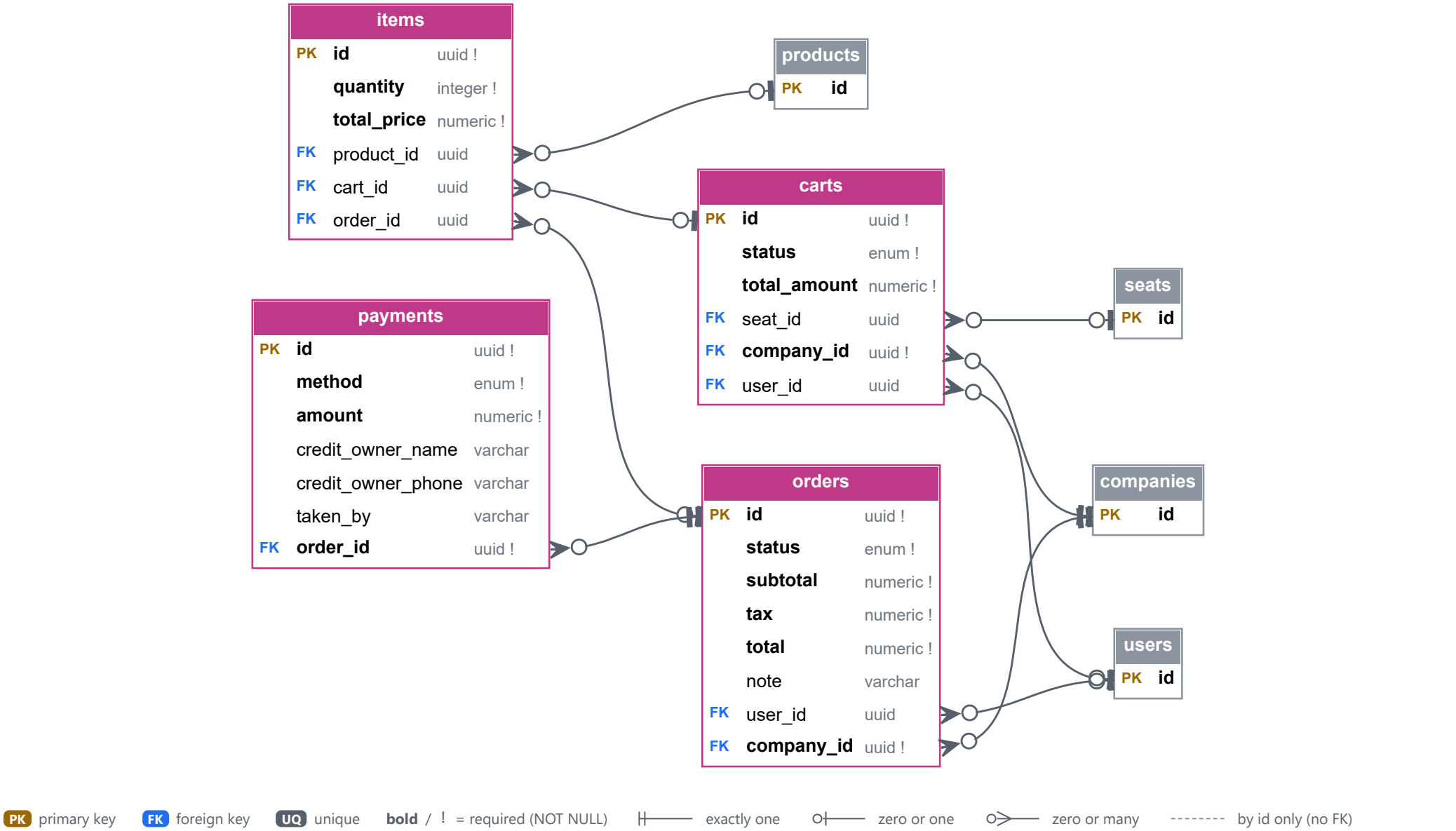
ERD: Products and catalog

What the company sells, and the images attached to products, categories, employees and users. Grey tables belong to another area (only their id is shown).



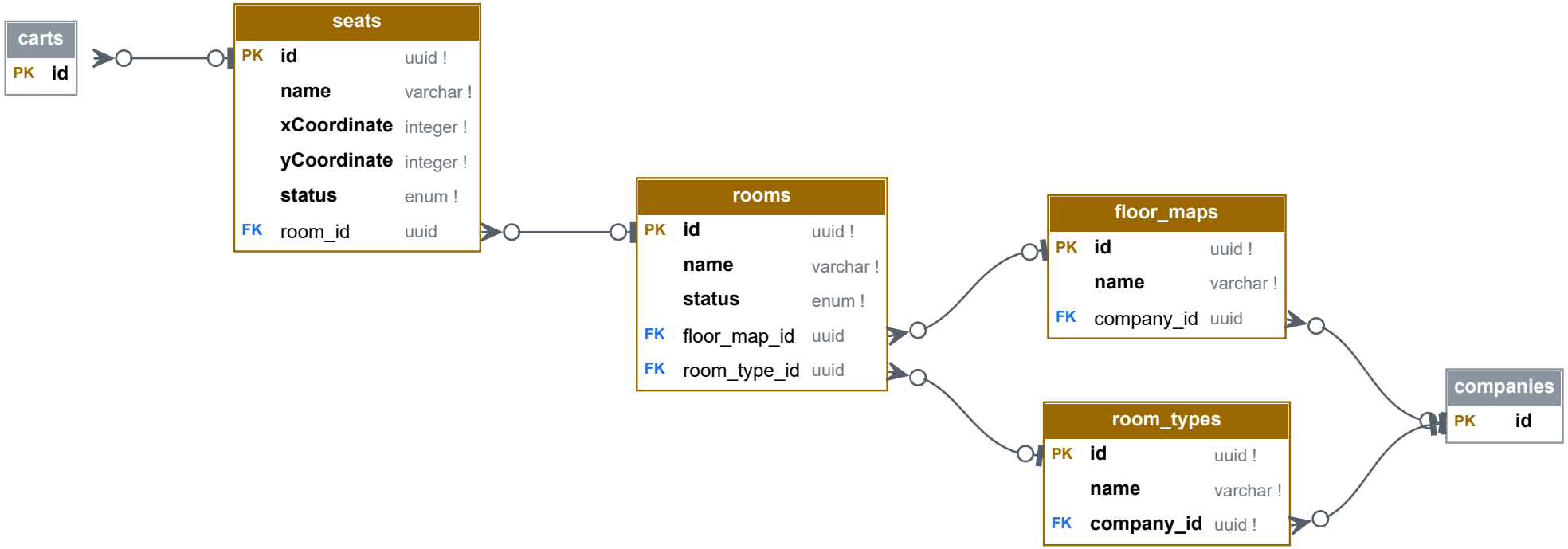
ERD: Sales: carts, orders and payments

A cart collects items on the till; checkout turns it into an order, which is then paid. Grey tables belong to another area (only their id is shown).



ERD: Floor plan

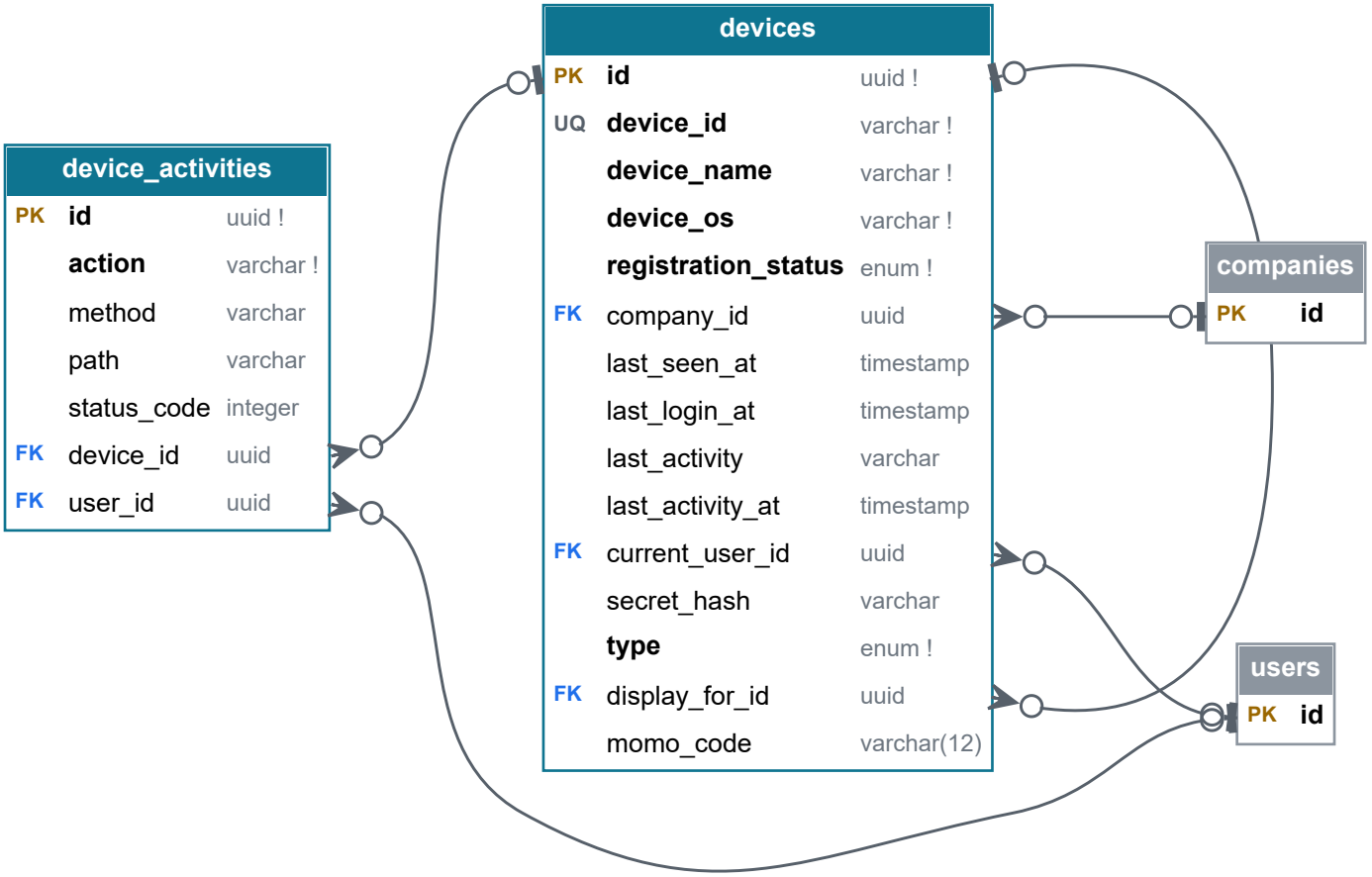
Floors, rooms and seats for restaurants and hotels. A cart can be opened for a seat. Grey tables belong to another area (only their id is shown).



PK primary key · FK foreign key · UQ unique · **bold** / ! = required (NOT NULL) · exactly one · zero or one · zero or many · by id only (no FK)

ERD: Devices

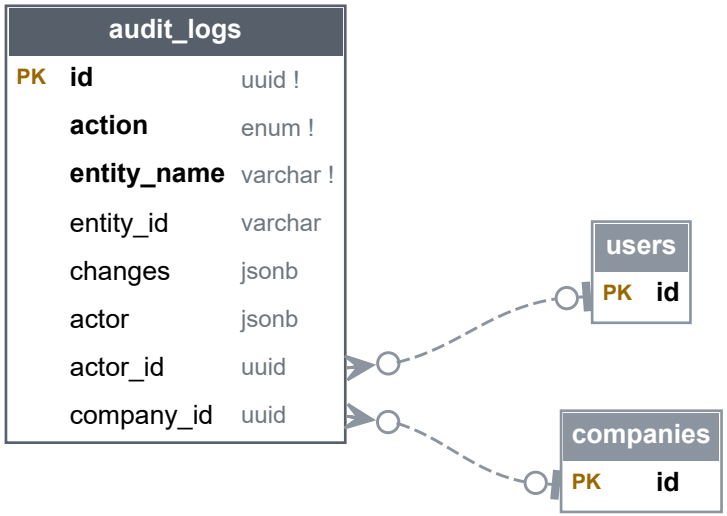
Registered terminals and their activity. A customer display points to the till whose cart it shows (devices.display_for_id). Grey tables belong to another area (only their id is shown).



PK primary key FK foreign key UQ unique **bold** / ! = required (NOT NULL) exactly one zero or one zero or many by id only (no FK)

ERD: History (audit log)

Every create / change / delete is recorded. The links to users and companies are by id only (dashed: not enforced by a foreign key), so history survives deletes. Grey tables belong to another area (only their id is shown).



PK primary key · FK foreign key · UQ unique · **bold** / ! = required (NOT NULL) · exactly one · zero or one · zero or many · by id only (no FK)

Relationships

All 31 foreign keys, read as "many rows of the left table point to one row of the right table". Required = the foreign key column cannot be empty. One-to-one = the foreign key column is unique.

Foreign key	Points to	Type	Required	On delete
carts.company_id	companies.id	many-to-one	**yes**	CASCADE
carts.seat_id	seats.id	many-to-one	no	NO ACTION
carts.user_id	users.id	many-to-one	no	SET NULL
device_activities.device_id	devices.id	many-to-one	no	CASCADE
device_activities.user_id	users.id	many-to-one	no	SET NULL
devices.company_id	companies.id	many-to-one	no	NO ACTION
devices.current_user_id	users.id	many-to-one	no	SET NULL
devices.display_for_id	devices.id	many-to-one	no	SET NULL
employee_roles.company_id	companies.id	many-to-one	no	NO ACTION
employees.company_id	companies.id	many-to-one	no	NO ACTION
employees.role_id	employee_roles.id	many-to-one	no	NO ACTION
files.category_id	product_categories.id	one-to-one	no	CASCADE
files.employee_id	employees.id	one-to-one	no	CASCADE
files.product_id	products.id	many-to-one	no	CASCADE
files.user_id	users.id	one-to-one	no	CASCADE
floor_maps.company_id	companies.id	many-to-one	no	NO ACTION
items.cart_id	carts.id	many-to-one	no	SET NULL
items.order_id	orders.id	many-to-one	no	NO ACTION
items.product_id	products.id	many-to-one	no	NO ACTION
orders.company_id	companies.id	many-to-one	**yes**	CASCADE
orders.user_id	users.id	many-to-one	no	NO ACTION
payments.order_id	orders.id	many-to-one	**yes**	CASCADE
product_categories.company_id	companies.id	many-to-one	**yes**	CASCADE
products.category_id	product_categories.id	many-to-one	no	NO ACTION
products.company_id	companies.id	many-to-one	**yes**	CASCADE

Foreign key	Points to	Type	Required	On delete
room_types.company_id	companies.id	many-to-one	**yes**	CASCADE
rooms.floor_map_id	floor_maps.id	many-to-one	no	NO ACTION
rooms.room_type_id	room_types.id	many-to-one	no	NO ACTION
seats.room_id	rooms.id	many-to-one	no	NO ACTION
users.company_id	companies.id	many-to-one	no	NO ACTION
users.employee_id	employees.id	one-to-one	no	NO ACTION
audit_logs.actor_id	users.id	many-to-one (by id only)	no	no FK
audit_logs.company_id	companies.id	many-to-one (by id only)	no	no FK

Enums

Fixed lists of values stored as PostgreSQL enum types.

Type	Values
audit_logs_action_enum	CREATE UPDATE DELETE
carts_status_enum	ACTIVE CHECKED_OUT
companies_type_enum	RESTAURANT BAR HOTEL SUPERMARKET PHARMACY SHOP
devices_registration_status_enum	PENDING REQUESTED REGISTERED REJECTED DISABLED
devices_type_enum	POS_TERMINAL KITCHEN_DISPLAY CUSTOMER_DISPLAY
employees_shift_enum	DAY EVENING NIGHT HYBRID
files_type_enum	image video raw auto
orders_status_enum	PENDING PREPARING READY COMPLETED CANCELLED
payments_method_enum	CASH MOMO CARD CREDIT
rooms_status_enum	AVAILABLE RESERVED OCCUPIED CLEANING MAINTENANCE
seats_status_enum	AVAILABLE RESERVED OCCUPIED CLEANING BLOCKED
users_role_enum	SUPERADMIN ADMIN OWNER EMPLOYEE

Table reference

Every column of every table. Bold "no" in Null = required.

audit_logs

History of every create / change / delete: who did it, when, and the before / after values.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	action	audit_logs_action (enum)	**no**		CREATE, UPDATE, DELETE
	entity_name	varchar	**no**		
	entity_id	varchar	yes		
	changes	jsonb	yes		Field: from / to
	actor	jsonb	yes		Snapshot of who did it
	actor_id	uuid	yes		User id (not a foreign key)
	company_id	uuid	yes		Company id (not a foreign key)

Indexes: (actor_id) · (company_id,createdAt) · (entity_id,entity_name)

carts

An open sale on a till. Items are added to it; at checkout it becomes an order and the cart is removed.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	status	carts_status (enum)	**no**		ACTIVE, CHECKED_OUT
	total_amount	numeric	**no**	0	Sum of its items, kept up to date
FK	seat_id	uuid	yes		→ seats.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK	company_id	uuid	**no**		→ companies.id (on delete cascade)
FK	user_id	uuid	yes		→ users.id (on delete set null)

Indexes: (company_id,status) · (user_id)

Referenced by: items.cart_id

companies

A business using the POS (restaurant, bar, hotel, supermarket, pharmacy, shop). Almost every other record belongs to one company.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
UQ	code	varchar	**no**		Code devices register with
	logo	varchar	yes		
	name	varchar	**no**		
	email	varchar	**no**		
	phone_number	varchar	**no**		
	location	varchar	**no**		
	type	companies_type (enum)	**no**		RESTAURANT, BAR, HOTEL, SUPERMARKET, PHARMACY, SHOP
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Indexes: unique (code)

Referenced by: carts.company_id, devices.company_id, employee_roles.company_id, employees.company_id, floor_maps.company_id, orders.company_id, product_categories.company_id, products.company_id, room_types.company_id, users.company_id

device_activities

Log of what was done on each device and by whom (login, requests).

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	action	varchar	**no**		
	method	varchar	yes		
	path	varchar	yes		

	Column	Type	Null	Default	Notes
	status_code	integer	yes		
FK	device_id	uuid	yes		→ devices.id (on delete cascade)
FK	user_id	uuid	yes		→ users.id (on delete set null)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Indexes: (createdAt,device_id)

devices

A terminal registered to a company: POS till, kitchen display or customer display. Owners approve it; it signs in with its device secret.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
UQ	device_id	varchar	**no**		Hardware id sent by the terminal
	device_name	varchar	**no**		
	device_os	varchar	**no**		
	registration_status	devices_registration_status (enum)	**no**	PENDING	PENDING, REQUESTED, REGISTERED, REJECTED, DISABLED
FK	company_id	uuid	yes		→ companies.id (on delete no action)
	last_seen_at	timestamp	yes		Online if seen in the last 2 minutes
	last_login_at	timestamp	yes		
	last_activity	varchar	yes		
	last_activity_at	timestamp	yes		
FK	current_user_id	uuid	yes		→ users.id (on delete set null). Who is signed in on it now
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
	secret_hash	varchar	yes		SHA-256 of the device secret
	type	devices_type (enum)	**no**	POS_TERMINAL	POS_TERMINAL, KITCHEN_DISPLAY, CUSTOMER_DISPLAY. What the device is used for
FK	display_for_id	uuid	yes		→ devices.id (on delete set null). Customer display: the till it shows
	momo_code	varchar(12)	yes		MoMo pay code (display falls back to its till)

Indexes: (display_for_id) · **unique** (device_id)

Referenced by: device_activities.device_id

employee_roles

A named set of permissions owners give to workers. permissions = NULL means everything; an empty list means nothing.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	description	varchar	yes		
FK	company_id	uuid	yes		→ companies.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
	permissions	text[]	yes		NULL = everything, [] = nothing

Referenced by: employees.role_id

employees

Staff record of a company (salary, shift, role). Linked one-to-one to the user account the worker signs in with.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	email	varchar	yes		
	phone	varchar	yes		
	salary	integer	**no**		
	shift	employees_shift (enum)	**no**		DAY, EVENING, NIGHT, HYBRID
FK	role_id	uuid	yes		→ employee_roles.id (on delete no action)
FK	company_id	uuid	yes		→ companies.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

	Column	Type	Null	Default	Notes
	extra_permissions	text[]	**no**	'{}'[]	Given to this worker on top of the role
	removed_permissions	text[]	**no**	'{}'[]	Taken away from this worker

Indexes: unique (company_id,email) · **unique** (company_id,phone)

Referenced by: files.employee_id, users.employee_id

files

Uploaded images (Cloudinary): product photos, category image, employee photo or user avatar.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	url	varchar	**no**		
	type	files_type (enum)	**no**		image, video, raw, auto
	mime_type	varchar	**no**		
	size	integer	**no**		
FK	product_id	uuid	yes		→ products.id (on delete cascade)
FK UQ	category_id	uuid	yes		→ product_categories.id (on delete cascade)
FK UQ	employee_id	uuid	yes		→ employees.id (on delete cascade)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK UQ	user_id	uuid	yes		→ users.id (on delete cascade)

Indexes: (product_id) · **unique** (category_id) · **unique** (employee_id) · **unique** (user_id)

floor_maps

A floor of the venue (restaurants, hotels).

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically

	Column	Type	Null	Default	Notes
	name	varchar	**no**		
FK	company_id	uuid	yes		→ companies.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Referenced by: rooms.floor_map_id

items

A product line (quantity and total price). Belongs to a cart while selling, then to the order after checkout.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	quantity	integer	**no**		
	total_price	numeric	**no**		
FK	product_id	uuid	yes		→ products.id (on delete no action)
FK	cart_id	uuid	yes		→ carts.id (on delete set null). Set while in a cart
FK	order_id	uuid	yes		→ orders.id (on delete no action). Set after checkout
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Indexes: (cart_id) · (order_id) · (product_id)

orders

A checked-out sale: PENDING, then PREPARING / READY (kitchen), then COMPLETED when paid, or CANCELLED.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	status	orders_status (enum)	**no**		PENDING, PREPARING, READY, COMPLETED, CANCELLED
	subtotal	numeric	**no**		
	tax	numeric	**no**		

	Column	Type	Null	Default	Notes
	total	numeric	**no**		subtotal + tax
	note	varchar	yes		
FK	user_id	uuid	yes		→ users.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK	company_id	uuid	**no**		→ companies.id (on delete cascade)

Indexes: (company_id,createdAt) · (user_id)

Referenced by: items.order_id, payments.order_id

payments

Money taken for an order. Split payments are several rows; CREDIT stores who owes it.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
	method	payments_method (enum)	**no**		CASH, MOMO, CARD, CREDIT
	amount	numeric	**no**		
	credit_owner_name	varchar	yes		CREDIT only: who owes
	credit_owner_phone	varchar	yes		CREDIT only
	taken_by	varchar	yes		
FK	order_id	uuid	**no**		→ orders.id (on delete cascade)

Indexes: (order_id)

product_categories

Groups products (names unique per company).

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically

	Column	Type	Null	Default	Notes
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK	company_id	uuid	**no**		→ companies.id (on delete cascade)

Indexes: **unique** (company_id,name)

Referenced by: files.category_id, products.category_id

products

Something the company sells, with buying and selling price and an optional barcode (unique per company).

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	buying_price	numeric	**no**		Never shown to customers
	selling_price	numeric	**no**		
	bar_code	varchar	yes		
	ingredients	text	yes		
	active	boolean	**no**	true	
FK	category_id	uuid	yes		→ product_categories.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK	company_id	uuid	**no**		→ companies.id (on delete cascade)

Indexes: (company_id) · **unique** (bar_code,company_id) · **unique** (company_id,name)

Referenced by: files.product_id, items.product_id

room_types

Kinds of rooms a company defines (e.g. VIP, Terrace, Double room).

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
FK	company_id	uuid	**no**		→ companies.id (on delete cascade)

Referenced by: rooms.room_type_id

rooms

A room or area on a floor map, with its type and status.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	status	rooms_status (enum)	**no**		AVAILABLE, RESERVED, OCCUPIED, CLEANING, MAINTENANCE
FK	floor_map_id	uuid	yes		→ floor_maps.id (on delete no action)
FK	room_type_id	uuid	yes		→ room_types.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Referenced by: seats.room_id

seats

A table or seat placed in a room (x / y position on the map). A cart can be opened for a seat.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	

	Column	Type	Null	Default	Notes
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
	xCoordinate	integer	**no**		
	yCoordinate	integer	**no**		
	status	seats_status (enum)	**no**		AVAILABLE, RESERVED, OCCUPIED, CLEANING, BLOCKED
FK	room_id	uuid	yes		→ rooms.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)
	updatedBy	jsonb	yes		Who made / last changed it (id, name, role)

Referenced by: carts.seat_id

users

Anyone who signs in: super admin, owner or employee. Employees sign in with a PIN on a device; one active session unless allowed several devices.

	Column	Type	Null	Default	Notes
PK	id	uuid	**no**	uuid_generate_v4()	
	createdAt	timestamp	**no**	now()	Set automatically
	updatedAt	timestamp	**no**	now()	Set automatically
	name	varchar	**no**		
UQ	email	varchar	yes		
	phone	varchar	yes		
	pin	varchar	yes		Employee sign-in PIN
	password	varchar	**no**		
	role	users_role (enum)	**no**		SUPERADMIN, ADMIN, OWNER, EMPLOYEE
	active	boolean	**no**	true	
	mustChangePassword	boolean	**no**	true	Forced password change on next login
	lastLoginAt	timestamp	yes		
FK UQ	employee_id	uuid	yes		→ employees.id (on delete no action)
FK	company_id	uuid	yes		→ companies.id (on delete no action)
	createdBy	jsonb	yes		Who made / last changed it (id, name, role)

Column	Type	Null	Default	Notes
updatedBy	jsonb	yes		Who made / last changed it (id, name, role)
allow_multiple_devices	boolean	**no**	false	Owner allows signing in on several devices at once
session_id	varchar	yes		The one session allowed (single-device login)

Indexes: **unique** (employee_id) · **unique** (email) · **unique** (company_id,email) · **unique** (company_id,phone)

Referenced by: carts.user_id, device_activities.user_id, devices.current_user_id, files.user_id, orders.user_id